NP-Completeness & Approximation Algorithms

Week 13 – Algorithms

39780: Algorithms, CLRS 3rd Edition Ch. 34 & 35

Part 1. Problem Complexity Classes

Motivation – An Analogy

Your boss assigns you an extremely hard problem to solve.

Scenario 1: "I can't find an efficient algorithm. I'm not smart enough."

Scenario 2: "I can't find an efficient algorithm, because no such algorithm exists."

Scenario 3 (NP-Complete): "I can't find an efficient algorithm, but neither could any of these famous experts."

> NP-Completeness theory lets you prove that a problem is as hard as thousands of other unsolved problems – giving you a *legitimate* reason to stop searching for an efficient solution.

Learning Objectives

- Distinguish between **P** and **NP**
- Understand the difference between **Decision** (Yes/No) problems and **Optimization** problems
- Define the meaning of **NP-Complete**
- Understand the **polynomial-time reduction** proof technique
- Recognize the practical benefits of proving a problem is NP-Complete

Classification of Problems

```
ALL PROBLEMS

Solvable (Decidable)

Tractable: Solvable in polynomial time
- Shortest Path, MST, Sorting, ...
- Time: O(n^k) for some constant k

Intractable: No known polynomial algorithm
*** NP-Complete problems live here ***
- SAT, TSP, Vertex Cover, Clique, ...
- Best known: O(2^n), O(n!), etc.

Super-exponential: e.g., Presburger arith.
```

Tractable Time = Polynomial Time

Polynomial time – input size n, running time bounded by $O(n^k)$ for constant k

Tractable (Polynomial)	Intractable (Super-polynomial)
$O(\log n)$	$O(2^n)$ – Exponential
$O(n)$	$O(n!)$ – Factorial
$O(n \log n)$	$O(n^n)$
$O(n^2), O(n^3)$	

Convention: A problem is "efficiently solvable" if and only if a polynomial-time algorithm exists for it.

Famous NP-Complete Problems – SAT

SAT (Satisfiability)

Given Boolean variables connected by OR ($\vee$) in clauses connected by AND ($\wedge$), find an assignment that satisfies **all** clauses.

Example: Variables w, x, y, z

$$(\overline{w} \vee x) \wedge (w \vee y) \wedge (\overline{x} \vee \overline{y} \vee z)$$

- **Solution:** $w = T, x = T, y = F, z = T$ or $z = F$

Another example (unsatisfiable):

$$(x) \wedge (\overline{x})$$

- **Solution:** None (no assignment can make both clauses true)

> SAT was the **first** problem proven NP-Complete (Cook, 1971).

Famous NP-Complete Problems – Subset Sum & Partition

Subset Sum

Given a set S of integers and a target K, does a subset of S sum to exactly K?

- $S = \{20, 30, 40, 80, 90\}$, $K = 200$
- **Solution:** $\{30, 80, 90\}$ – sum is 200

Partition

Given a set S of integers, can S be divided into two subsets with equal sum?

- $S = \{20, 30, 40, 80, 90\}$ (total = 260)
- **Solution:** $X = \{20, 30, 80\}$, $Y = \{40, 90\}$ – both sum to 130

> Subset Sum can be reduced to Partition in polynomial time.

Famous NP-Complete Problems – 0-1 Knapsack

0-1 Knapsack (Decision Version)

Capacity C, items with weights w_i and values v_i. Is there a subset with total weight $\leq C$ and total value $\geq K$?

Example: $C = 20\text{kg}$

Item	Weight	Value
1	12 kg	20
2	8 kg	10
3	6 kg	15
4	5 kg	25

Famous NP-Complete Problems – Graph Problems

Vertex Cover

Find the **smallest** set of vertices such that every edge has at least one endpoint in the set.

Independent Set

Find the **largest** set of vertices such that no two are adjacent.

Clique

Find the **largest** complete subgraph (every pair of vertices is connected).

> **Relationship:** In graph $G = (V, E)$, if S is a vertex cover, then $V \setminus S$ is an independent set (and vice versa). A clique in G corresponds to an independent set in the complement graph $\overline{G}$.

Famous NP-Complete Problems – More Graph Problems

Graph Coloring

Color vertices so no two adjacent vertices share a color. Minimize the number of colors.

Longest Path

Given weighted graph G, vertices s and t, and constant K: does a simple path from s to t of length $\geq K$ exist?

> **Counterintuitive:** Shortest Path is in **P** (Dijkstra: $O(V^2)$), but Longest Path is **NP-Complete**!

Set Cover

Given set S and a collection of subsets, find the fewest subsets whose union equals S.

Famous NP-Complete Problems – TSP & Friends

Traveling Salesman Problem (TSP)

Given a weighted complete graph, find the shortest Hamiltonian cycle (visit every vertex exactly once and return).

Hamiltonian Cycle

Does a given graph have a cycle that visits every vertex exactly once?

- TSP with all edge weights equal reduces to Hamiltonian Cycle

Bin Packing

Given n items and bins of capacity C, pack all items using the fewest bins.

Decision Problems vs Optimization Problems

Aspect	Decision Problem	Optimization Problem
Answer	Yes or No	The best solution
Example (TSP)	"Is there a tour of length $\leq K$?"	"What is the shortest tour?"
Example (Shortest Path)	"Is there a path of length ≤ 100?"	"What is the shortest path length?"

Key insight – they are two sides of the same coin:

- If the **decision** version is hard, the **optimization** version is **at least as hard**
- NP-Completeness theory focuses on **decision problems**
- But hardness of the decision version implies hardness of the optimization version

Part 2. P, NP, and NP-Completeness Theory

The Class P

Definition

$\mathbf{P}$ = the set of decision problems solvable by a **deterministic** algorithm in **polynomial time**.

- Given a question, the algorithm outputs **Yes** or **No** in $O(n^k)$ time

Examples in P:

- Is a graph connected? – BFS/DFS: $O(V + E)$
- Is there a shortest path from u to v of length $\leq K$? – Dijkstra: $O(V^2)$
- Can we sort n numbers? – Merge Sort: $O(n \log n)$
- Is there a minimum spanning tree of weight $\leq K$? – Kruskal: $O(E \log E)$

> All the algorithms we studied this semester solve problems in **P**.

The Class NP

Definition

$\mathbf{NP}$ = **Nondeterministic Polynomial** time

A decision problem is in NP if: given a **"Yes" certificate** (a proposed solution), we can **verify** it in polynomial time.

> **Important:** NP does NOT mean "Non-Polynomial"!

Intuitive definition:

- **P** = problems we can **solve** quickly
- **NP** = problems we can **verify** quickly

NP – Examples

Hamiltonian Cycle is in NP:

- Certificate: a sequence of vertices $(v_1, v_2, \ldots, v_n, v_1)$
- Verification: check that all vertices appear exactly once, and each consecutive pair is connected by an edge
- Time: $O(n)$ – polynomial!

SAT is in NP:

- Certificate: an assignment of True/False to each variable
- Verification: substitute values and check if all clauses are satisfied
- Time: $O(m \cdot k)$ where m = number of clauses, k = variables per clause

Subset Sum is in NP:

- Certificate: a subset of integers
- Verification: check that they sum to K

P vs NP – The Relationship

```
Possibility (a): P ≠ NP          Possibility (b): P = NP
     (Strongly believed)              (Would be shocking)

 ┌──────────────────────┐        ┌──────────────────────┐
 │          NP          │        │                      │
 │   ┌──────────────┐   │        │        P = NP        │
 │   │              │   │        │                      │
 │   │      P       │   │        │  Everything in NP    │
 │   │              │   │        │  would be solvable   │
 │   │              │   │        │  in polynomial time  │
 │   │              │   │        │                      │
 │   │              │   │        │                      │
 │   │              │   │        │                      
```

Polynomial-Time Reduction

Core Mechanism of NP-Completeness Theory

Problem A **reduces to** problem B (written $A \leq_P B$) if:

1. Every instance of A can be **transformed** into an instance of B in **polynomial time**
2. The **Yes/No answers are preserved**: instance α of A is Yes $\iff$ transformed instance β of B is Yes

```
                    Polynomial-time
Instance alpha  ------------------->  Instance beta
(Problem A)          transform         (Problem B)
     |                                      |
     |                                      ▼
     |                                 Algorithm for B
     |                                      |
     ▼                                      ▼
  Answer:                                Answer:
  Yes  <------------------------------>  Yes
  No   <------------------------------>  No
```

Consequence: If $A \leq_P B$ and B is solvable in polynomial time, then A is also solvable in polynomial time.

Reduction Example 1 – Simple Number Theory

Problem 1: Is integer $x = x_1 x_2 \ldots x_n$ (digit representation) divisible by 3?

Problem 2: Is $x_1 + x_2 + \ldots + x_n$ divisible by 3?

These two problems have **identical answers** for every input (a well-known divisibility rule).

- Transformation: extract digits and sum them – $O(n)$ time
- Problem 2 is easy (just compute a sum and check mod 3)
- Therefore Problem 1 is also easy

$$\text{Problem 1} \leq_P \text{Problem 2}$$

Reduction Example 2 – HAM-CYCLE to TSP

HAM-CYCLE: Given graph $G = (V, E)$, does G have a Hamiltonian cycle?

TSP (decision): Given weighted complete graph G' and bound K, does G' have a Hamiltonian cycle of total weight $\leq K$?

Reduction: HAM-CYCLE $\leq_P$ TSP

Transform instance of HAM-CYCLE into instance of TSP:

```
HAM-CYCLE instance          TSP instance
(4 vertices, 5 edges)       (complete graph, all 6 edges)

  a --- b                    a --1-- b
  |   / |        ⟹          |1\  /1 |1
  |  /  |                    |  ∨    |
  c --- d                    c -1/∞- d

Existing edge  ⟶ weight 1       Set K = |V| = 4
Missing edge   ⟶ weight ∞
```

NP-Hard – Definition

Definition (NP-Hard)

A problem A is **NP-Hard** if:

$$\text{For every problem } L \in NP: \quad L \leq_P A$$

In other words, A is **at least as hard** as every problem in NP.

Theorem 13-1 (Practical NP-Hard Proof)

To prove A is NP-Hard, it suffices to show:

$$\text{Some known NP-Hard problem } C \leq_P A$$

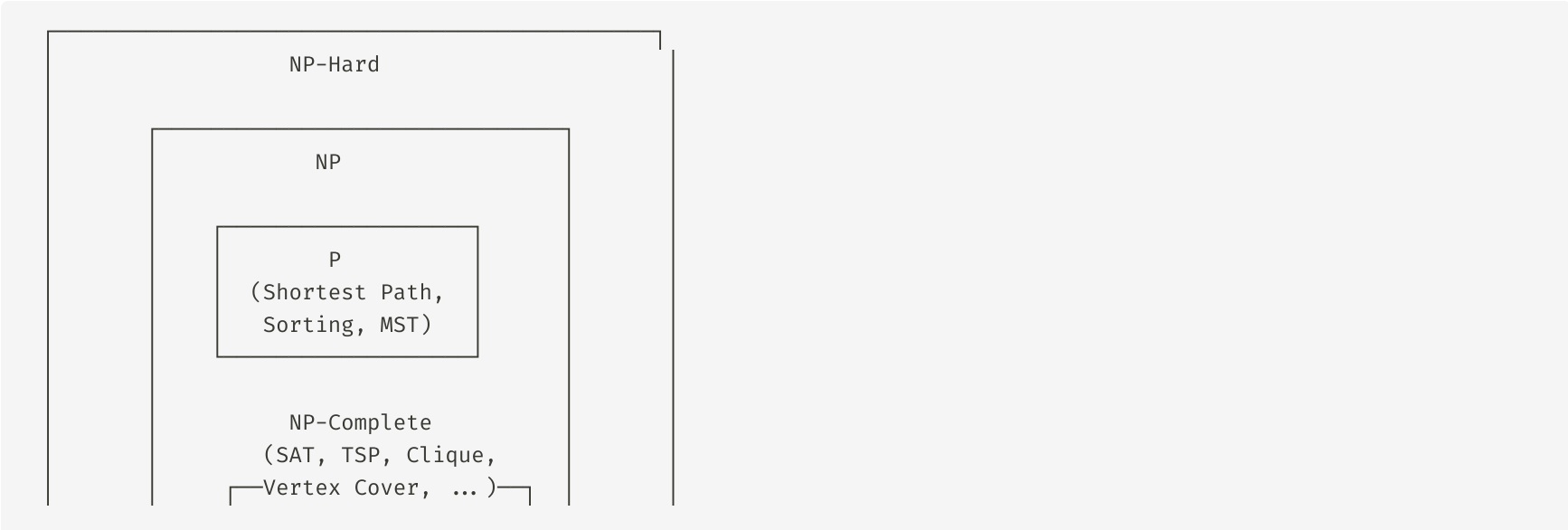
Why this works: If every NP problem reduces to C, and C reduces to A, then every NP problem reduces to A (by transitivity of polynomial-time reductions).

NP-Complete – Definition

Definition (NP-Complete)

A problem A is **NP-Complete** if:

1. $A \in NP$ (we can verify a "Yes" certificate in polynomial time)
2. A is **NP-Hard** (every NP problem reduces to A)



How to Prove NP-Completeness

Two-Step Recipe

Step 1: Show that problem A is in **NP**

- Describe a polynomial-time verification algorithm for a "Yes" certificate
- Usually straightforward

Step 2: Show that problem A is **NP-Hard**

- Pick a known NP-Hard (or NP-Complete) problem C
- Construct a polynomial-time reduction $C \leq_P A$
- Prove the Yes/No answers are preserved in both directions

```
Polynomial-time reduction
  alpha ———————————————————> beta
  (Problem C instance)       (Problem A instance)
        |                          |
        |                     Algorithm for A
        |                          |
        ▼                          ▼
```

NP-Completeness Proof – Longest Path

Definitions

HAM-PATH-2-POINTS: Given $G = (V, E)$ and vertices s, t, does a Hamiltonian path from s to t exist? *(Known NP-Complete)*

LONGEST-PATH: Given weighted $G = (V, E)$, vertices s, t, and constant K, does a simple path from s to t of length $\geq K$ exist?

Proof: LONGEST-PATH is NP-Complete

Step 1: LONGEST-PATH $\in$ NP

- **Certificate:** A sequence of vertices forming a path from s to t
- **Verification:** Walk along the path, check all edges exist in E, sum the weights, confirm $\geq K$
- **Time:** $O(|V|)$ – polynomial

Step 2: HAM-PATH-2-POINTS $\leq_P$ LONGEST-PATH

Transformation: Given HAM-PATH instance $G = (V, E)$ with endpoints s, t:

- Copy G exactly, assign weight **1** to every edge
- Set $K = |V| - 1$

```
HAM-PATH-2-POINTS instance       LONGEST-PATH instance

s --- a --- b                     s -1- a -1- b
|           |          ⟹          |           |
c --- d --- t                     c -1- d -1- t
```

The Reduction Chain — History

The first NP-Complete proof was for **SAT** (Boolean Satisfiability) by Cook (1971), proved directly from the definition.

All subsequent proofs use **reduction from a known NP-Complete problem**:

```
CIRCUIT-SAT (Cook-Levin, 1971)
  │
  ▼
 SAT
  │
  ▼
3-SAT ──────────────────────────────┐
  │                                 │
  ├──────────┐                      │
  ▼          ▼                      ▼
CLIQUE   SUBSET-SUM            VERTEX-COVER
  │                                 │
  │                       ┌─────────┼─────────┐
  │                       ▼         ▼         ▼
  │                  HAM-CYCLE HAM-PATH HAM-PATH-2-PTS
  │                      │                   │
  │                      ▼                   ▼
  │                     TSP            LONGEST-PATH
```

Counterintuitive Example

Some problems that **look** similar have vastly different complexity:

Problem	Complexity	Algorithm
Shortest Path	P	Dijkstra $O(V^2)$
Longest Path	NP-Complete	No known poly-time algorithm
Euler Circuit (visit every **edge** once)	P	Hierholzer $O(E)$
Hamiltonian Cycle (visit every **vertex** once)	NP-Complete	No known poly-time algorithm
2-SAT	P	Implication graph $O(n+m)$
3-SAT	NP-Complete	No known poly-time algorithm
2-Coloring	P	BFS/DFS $O(V+E)$

What Does "NP-Complete" Mean in Practice?

When your problem is proven NP-Complete:

1. **Stop** searching for an exact polynomial-time algorithm – no one has found one in 50+ years
2. **Use heuristics** – algorithms that find good (but not necessarily optimal) solutions quickly
3. **Use approximation algorithms** – algorithms with provable bounds on solution quality
4. **Use special-case solvers** – many NP-Complete problems have polynomial-time solutions for restricted inputs

> *"Sometimes it is useful to know that something is impossible."* – **Leonid Levin** (co-discoverer of NP-Completeness, independently of Cook)

The Venn Diagram — P, NP, NP-Hard, NP-Complete

```
Assuming P ≠ NP (widely believed):

┌──────────────────────────────────────────────────────┐
│                      NP-Hard                         │
│                                                      │
│   ┌──────────────────────────────────────────┐       │
│   │                    NP                    │       │
│   │                                          │       │
│   │   ┌──────────────────────┐               │       │
│   │   │                      │               │       │
│   │   │          P           │  NP-Complete  │       │
│   │   │                      │               │       │
│   │   │  - Sorting           │  ┌──────────┐ │       │
│   │   │  - Shortest Path     │  │ SAT      │ │       │
│   │   │  - MST               │  │ TSP      │ │       │
│   │   │  - Matching          │  │ Clique   │ │       │
│   │   │                      │  │ V-Cover  │ │       │
│   │   │                      │  │ 3-SAT    │ │       │
│   │   │                      │  │ SubsetSum│ │       │
│   │   │                      │  └──────────┘ │       │
│   │   └──────────────────────┘               │       │
│   │                                          │       │
│   └──────────────────────────────────────────┘       │
│                                                      │
│   NP-Hard but NOT in NP:                             │
│   - Halting Problem                                  │
└──────────────────────────────────────────────────────┘
```

3-SAT to CLIQUE Reduction – Sketch

3-SAT: Given clauses each with exactly 3 literals, is there a satisfying assignment?

CLIQUE: Does graph G have a complete subgraph of size k?

Reduction: 3-SAT $\leq_P$ CLIQUE

Given 3-SAT instance with m clauses, construct a graph:

- For each literal in each clause, create a vertex (so $3m$ vertices total)
- Connect two vertices with an edge **unless**:
  - They are in the **same clause**, OR
  - They are **contradictory** (e.g., x_1 and $\overline{x_1}$)
- Set $k = m$ (number of clauses)

Example: $(x_1 \vee x_2 \vee x_3) \wedge (\overline{x_1} \vee x_2 \vee x_3)$

Summary

Concept	Definition
P	Problems solvable in polynomial time
NP	Problems whose "Yes" answer is *verifiable* in polynomial time
NP-Hard	At least as hard as every problem in NP
NP-Complete	In NP AND NP-Hard
Reduction $A \leq_P B$	Transform A to B in poly-time, preserving answers

Key takeaways:

1. $P \subseteq NP$ (trivially). Whether $P = NP$ is unknown.
2. To prove NP-Completeness: (a) show problem is in NP, (b) reduce a known NP-Complete problem to it.

Part 3. Approximation Algorithms

Why Approximation Algorithms?

NP-complete problems appear everywhere in the real world, yet no polynomial-time algorithm has been found for any of them.

To deal with NP-complete problems, **we must give up one** of the following three:

1. Finding a solution **in polynomial time**
2. Finding a solution **for all inputs**
3. Finding the **optimal** solution

Approximation algorithms give up (3) – they find a near-optimal solution in polynomial time.

Approximation Ratio

An approximation algorithm must come with an **approximation ratio** that tells us how close the approximate solution is to the optimal solution.

$$\text{Approximation Ratio} = \frac{\text{Approximate Solution Value}}{\text{Optimal Solution Value}}$$

- A ratio closer to **1.0** means higher accuracy
- **Problem**: Computing the ratio requires knowing the optimal solution (which is what we cannot find!)
- **Solution**: Use an **indirect optimal solution** – a value we can compute that is guaranteed to be a bound on the true optimal

Indirect Optimal Solution – Key Idea

For **minimization** problems:

- Find a computable value **L** such that **OPT >= L** (lower bound)
- Show that the approximate solution **APX <= c * L**
- Then **APX <= c * OPT**, giving approximation ratio **c**

For **maximization** problems:

- Find a computable value **U** such that **OPT <= U** (upper bound)
- Show that the approximate solution **APX >= U / c**
- Then **APX >= OPT / c**

1. Traveling Salesman Problem (TSP)

TSP – Problem Definition

Traveling Salesman Problem (TSP)

- A salesperson starts from a city, visits every other city exactly once, and returns to the starting city
- **Goal**: Minimize the total travel distance

Conditions (Metric TSP):

- **Symmetry**: distance(A, B) = distance(B, A)
- **Triangle inequality**: distance(A, B) <= distance(A, C) + distance(C, B)

These conditions hold for Euclidean distances and many practical scenarios.

MST-Based Approach – Intuition

To design a polynomial-time approximation, find a related problem with a known efficient algorithm:

Minimum Spanning Tree (MST) shares key properties with TSP:

- MST connects **all vertices** (like TSP visits all cities)
- MST minimizes **total edge weight** (related to minimizing tour length)

We leverage MST to construct an approximate TSP tour.

Approx_MST_TSP – Algorithm

Input: n cities with pairwise distances

Output: A tour visiting each city exactly once and returning to the start

1. Compute the **MST** of the input graph (using Kruskal's or Prim's)
2. Starting from any city, perform a **DFS traversal** of the MST, recording the visit order (each edge traversed twice)
3. **Remove duplicate cities** from the visit order (keep the first occurrence of each city, plus the starting city at the end)

Return the resulting tour

Worked Example – Step 1: Build MST

Consider 8 cities: **A, B, C, D, E, F, G, H**

```
Original Graph          MST (computed by Kruskal/Prim)

A --- B --- C               A
|   / |   / |              / \
|  /  |  /  |             G   (other edges)
| /   | /   |            / \
D --- E --- F            E   D
|         / |              / \
|        /  |             H   F
G ------H   |            / \   \
             |           C   B   ...
```

Apply Kruskal's or Prim's algorithm to find the MST.

Worked Example – Step 2: DFS Traversal

Starting from city **A**, traverse the MST edges (each edge used exactly twice):

Full traversal order (with backtracking):

$$A \to G \to E \to G \to D \to H \to C \to H \to D \to F \to B \to F \to D \to G \to A$$

Total distance of this traversal = **2M** (where M = total MST weight)

because each of the (n-1) tree edges is traversed exactly **twice** (once forward, once back).

Worked Example – Step 3: Remove Duplicates

Apply triangle inequality to shortcut through already-visited cities:

$$A \to G \to E \to \cancel{G} \to D \to H \to C \to \cancel{H} \to \cancel{D} \to F \to B \to \cancel{F} \to \cancel{D} \to \cancel{G} \to A$$

Resulting tour: A -> G -> E -> D -> H -> C -> F -> B -> A

By the **triangle inequality**, each shortcut is no longer than the path it replaces.

Therefore, the tour length **<= 2M**.

TSP – Approximation Ratio Proof

Claim: Approx_MST_TSP has approximation ratio **<= 2.0**

Proof:

Let **M** = total weight of MST, **OPT** = optimal TSP tour length, **APX** = approximate tour length.

1. **OPT > M**: The optimal tour visits all cities and returns to start. Removing any edge from the optimal tour gives a spanning tree. Since MST is the minimum spanning tree, M <= (tour minus one edge) < OPT.

2. **APX <= 2M**: The DFS traversal has length exactly 2M. Shortcutting via triangle inequality can only decrease the length.

3. **Combining**: APX <= 2M < 2 * OPT

$$\frac{APX}{OPT} < 2.0$$

TSP – Time Complexity

Step	Operation	Complexity
1	Build MST (Kruskal's)	O(m log m)
1	Build MST (Prim's)	O(n^2)
2	DFS traversal of MST	O(n)
3	Remove duplicates	O(n)

Overall: Dominated by MST construction

- Kruskal's: **O(m log m)** where m = number of edges
- Prim's: **O(n^2)** where n = number of cities

2. Vertex Cover Problem

Vertex Cover – Problem Definition

Vertex Cover of a graph G = (V, E):

- A subset S of V such that **every edge** in E has at least one endpoint in S
- **Goal**: Find the **minimum-size** vertex cover

Real-world analogy: Placing the minimum number of CCTV cameras at intersections so that every corridor (edge) is monitored.

Example: For a triangle graph {1, 2, 3}:

- {1, 2, 3}, {1, 2}, {1, 3}, {2, 3} are all vertex covers
- {1} is the minimum vertex cover if vertex 1 connects to all edges
- {2} alone fails if edge (1,3) is uncovered

Maximal Matching Approach

Key insight: Instead of the expensive set cover reduction, use **maximal matching**.

Matching: A set of edges with no shared endpoints

Maximal Matching: A matching where no more edges can be added

- Not the same as *maximum* matching (which is the largest possible)
- A maximal matching can be found greedily in polynomial time

Idea: Select edges greedily (if neither endpoint is already covered), then take all endpoints as the vertex cover.

Approx_Matching_VC – Algorithm

Input: Graph G = (V, E)

Output: A vertex cover

1. Find a **maximal matching** M in G:
   - For each edge (u, v) in E:
     - If neither u nor v is an endpoint of an already-selected edge, add (u, v) to M
2. **Return** the set of all endpoints of edges in M

Worked Example – Vertex Cover

```
Given graph with 9 vertices:
   1 --- 2 --- 3
   |   / |   / |
   4 --- 5 --- 6
   |   / |   / |
   7 --- 8 --- 9
```

Maximal matching M (selected edges shown): edges a, b, c, d, e, f

- Approximate solution: **12 endpoints** (all endpoints of 6 matching edges)
- Optimal solution: **7 vertices**
- Ratio: 12/7 ~ 1.71 (within the guaranteed bound of 2.0)

Vertex Cover – Approximation Ratio Proof

Claim: Approx_Matching_VC has approximation ratio **= 2.0**

Proof:

Let |M| = number of edges in the maximal matching, OPT = size of optimal vertex cover.

1. **OPT >= |M|**: Any vertex cover must include at least one endpoint of each matching edge. Since matching edges share no endpoints, at least |M| distinct vertices are needed.

2. **APX = 2|M|**: The algorithm returns both endpoints of every matching edge.

3. **Combining**:

$$\frac{APX}{OPT} = \frac{2|M|}{OPT} \leq \frac{2|M|}{|M|} = 2.0$$

Vertex Cover – Time Complexity

Finding a maximal matching:

- For each edge, check if its endpoints are already used: **O(n)** per edge
- Total edges: m

Time Complexity: **O(nm)**

3. Bin Packing Problem

Bin Packing – Problem Definition

Bin Packing Problem:

- Given n items with sizes s_1, s_2, …, s_n and bins of capacity **C**
- Each item size <= C
- **Goal**: Pack all items into the **fewest number of bins**

This is an NP-hard optimization problem with many practical applications (memory allocation, container loading, disk partitioning).

Four Greedy Heuristics

Method	Strategy
First Fit (FF)	Scan bins from the first; place item in the **first bin with enough room**
Next Fit (NF)	Check only the **most recently used bin**; if it fits, place it there; otherwise open a new bin
Best Fit (BF)	Place item in the bin where it fits with the **least remaining space**
Worst Fit (WF)	Place item in the bin where it fits with the **most remaining space**

For all four methods: if no existing bin has room, **open a new bin**.

Worked Example – Bin Packing

C = 10, items = [7, 5, 6, 4, 2, 3, 7, 5]

Method	Bin Contents	Bins Used
First Fit	{7,3}, {5,4}, {6,2}, {7}, {5}	5
Next Fit	{7}, {5}, {6,4}, {2,3}, {7}, {5}	6
Best Fit	{7,3}, {5,4}, {6,2}, {7}, {5}	5
Worst Fit	{7,2}, {5,4}, {6,3}, {7}, {5}	5
Optimal	{7,3}, {5,5}, {6,4}, {7,2}	**4**

Approx_BinPacking – Algorithm

Input: n items with sizes s_1, …, s_n

Output: Number of bins used B

```
B = 0
for i = 1 to n:
    if there exists a bin with enough room (by greedy strategy):
        place item i in that bin
    else:
        open a new bin
        place item i in the new bin
        B = B + 1
return B
```

Bin Packing – Time Complexity

Method	Per-item work	Total
First Fit	Scan all bins: O(n)	**O(n^2)**
Best Fit	Scan all bins: O(n)	**O(n^2)**
Worst Fit	Scan all bins: O(n)	**O(n^2)**
Next Fit	Check one bin: O(1)	**O(n)**

Bin Packing – Ratio Proof (FF, BF, WF)

Claim: First Fit, Best Fit, and Worst Fit all have approximation ratio **<= 2.0**

Key observation: At most **one bin** can be less than half full.

- If two bins were each less than half full, their items could fit in one bin, contradicting the greedy strategy.

Proof:

Let OPT = optimal number of bins, OPT′ = number of bins used by the algorithm.

- OPT >= (sum of all item sizes) / C
- At least (OPT′ - 1) bins are more than half full:

$$\sum s_i > (OPT' - 1) \times \frac{C}{2}$$

$$\frac{\sum s_i}{C} > \frac{OPT' - 1}{2} \implies OPT > \frac{OPT' - 1}{2}$$

$$2 \cdot OPT > OPT' - 1 \implies 2 \cdot OPT + 1 > OPT' \implies 2 \cdot OPT \geq OPT'$$

Bin Packing – Ratio Proof (Next Fit)

Claim: Next Fit has approximation ratio **<= 2.0**

Key observation: For any two consecutive bins, their combined item sizes exceed C.

- Otherwise, the items in the second bin would have fit in the first.

Proof:

Consider consecutive bin pairs: (bin 1, bin 2), (bin 3, bin 4), …

Each pair has total item size > C. There are at least OPT′/2 such pairs.

$$\sum s_i > \frac{OPT'}{2} \times C$$

$$\frac{\sum s_i}{C} > \frac{OPT'}{2} \implies OPT > \frac{OPT'}{2} \implies 2 \cdot OPT > OPT'$$

4. Job Scheduling Problem

Job Scheduling – Problem Definition

Job Scheduling Problem:

- n jobs with processing times $t_1, t_2, \ldots, t_n$
- m identical machines $M_1, M_2, \ldots, M_m$
- Each job runs on exactly one machine without interruption
- Each machine processes one job at a time

Goal: Assign jobs to machines to **minimize the makespan** (the time when the last job finishes).

Approx_JobScheduling – Algorithm

Input: n jobs with times t_i, m machines

Output: Makespan (latest finishing time)

```
for j = 1 to m:
    L[j] = 0        // finishing time of machine j

for i = 1 to n:
    min = 1
    for j = 2 to m:                  // find earliest-finishing machine
        if L[j] < L[min]:
            min = j
    assign job i to machine M_min
    L[min] = L[min] + t_i

return max(L[1], ... , L[m])
```

Strategy: Always assign the next job to the machine that finishes earliest.

Worked Example – Job Scheduling

Jobs: t = [5, 2, 4, 3, 4, 7, 9, 2, 4, 1], **4 machines**

Step-by-step assignment:

Job	t_i	Assigned to	Machine loads after
1	5	M1	[5, 0, 0, 0]
2	2	M2	[5, 2, 0, 0]
3	4	M3	[5, 2, 4, 0]
4	3	M4	[5, 2, 4, 3]
5	4	M2	[5, 6, 4, 3]
6	7	M4	[5, 6, 4, 10]

Job Scheduling – Approximation Ratio Proof

Claim: Approx_JobScheduling has approximation ratio **<= 2.0**

Let the last job assigned be job i, starting at time T. Then OPT′ = T + t_i.

Two lower bounds on OPT:

1. **OPT >= (sum of all t_i) / m**: Even with perfect load balancing, no machine finishes before the average.

2. **OPT >= t_i**: The optimal solution must also process job i, taking at least t_i time.

Proof:

$$T \leq T' = \frac{\sum_{j \neq i} t_j}{m} \leq \frac{\sum t_j}{m} \leq OPT$$

T is when the earliest machine was free; T′ is the average excluding job i. Since job i went to the earliest machine, T <= T′.

$$OPT' = T + t_i \leq OPT + OPT = 2 \cdot OPT$$

Job Scheduling – Time Complexity

- For each of n jobs, find the minimum among m machines: O(m)
- Finding the maximum at the end: O(m)

Time Complexity: n * O(m) + O(m) = **O(nm)**

5. Clustering Problem

Clustering – Problem Definition

k-Clustering Problem:

- Given n points in a 2D plane and an integer k
- Partition the points into **k groups**, each with a designated **center** point
- **Goal**: Minimize the **maximum group diameter** (the diameter of the largest cluster)

Applications: Recommendation systems, data mining, VLSI design, parallel processing, web search, pattern recognition, gene analysis, social network analysis, and many more.

Farthest-First Traversal – Greedy Strategy

Idea: Select centers one at a time, each time choosing the point **farthest from all existing centers**.

1. Pick an arbitrary point as the first center C_1
2. For the j-th center (j = 2, …, k):
   - For each non-center point x_i, compute D[i] = distance to the **nearest** existing center
   - Select the point with the **largest** D[i] as the next center C_j
3. Assign each non-center point to its nearest center

Intuition: Spreading centers far apart ensures each cluster covers a small area.

Approx_k_Clusters – Algorithm

Input: n points x_0, …, x_{n-1}, number of groups k > 1

Output: k groups with centers

```
C[1] = x_r                      // random first center
for j = 2 to k:
    for i = 0 to n-1:
        if x_i is not a center:
            D[i] = min distance from x_i to any existing center
    C[j] = x_i where D[i] is maximum (and x_i is not a center)

Assign each non-center point to its nearest center
return C and cluster assignments
```

Worked Example – Clustering (k = 4)

Step 1: Pick arbitrary point as C_1

Step 2: Compute D[i] for all points; farthest from C_1 becomes C_2

Step 3: For each point, D[i] = min(dist(x_i, C_1), dist(x_i, C_2))

- Example: D[1]=18, D[2]=19, D[3]=20, D[4]=17 (others < 20)
- Point with D[3]=20 (largest) becomes C_3

Step 4: Recompute D[i] = min(dist to C_1, C_2, C_3); farthest becomes C_4

Step 5: Assign each remaining point to its nearest center

Clustering – Time Complexity

Component	Analysis
Inner for-loop (line 3–5)	Each point computes distance to all centers: O(kn)
Find max D[i] (line 6)	O(n)
Outer for-loop (line 2)	Repeats (k-1) times
Final assignment (line 7)	O(kn)

Total: (k-1) * (O(kn) + O(n)) + O(kn) = **O(k^2 n)**

Clustering – Approximation Ratio Proof

Claim: Approx_k_Clusters has approximation ratio **<= 2.0**

Setup: After finding k centers, imagine selecting a **(k+1)-th virtual center** C_{k+1} using the same farthest-first rule. Let **d** = distance from C_{k+1} to its nearest center.

Step 1: OPT >= d

- We now have (k+1) center points that must be divided into k groups
- By pigeonhole principle, at least two centers share a group
- The diameter of that group >= d, so OPT >= d

Step 2: OPT' <= 2d

- C_{k+1} is the farthest point from any center, so every other point is within distance d of its nearest center
- Each cluster has radius <= d, so diameter <= 2d
- Therefore OPT' <= 2d

Conclusion: 2 * OPT >= 2d >= OPT', so approximation ratio <= **2.0**

Clustering – Practical Considerations

Random first center: Since C_1 is chosen randomly, results vary between runs.

- Run the algorithm multiple times and take the **best result**

Outlier sensitivity: Noisy data or outliers can distort center selection.

- **Preprocess** data to remove outliers before applying the algorithm

Note: This algorithm is closely related to the **k-center problem** and has inspired practical algorithms like **k-means++** initialization.

Approximation Algorithms – Summary Table

Problem	Indirect Optimal	Ratio	Time Complexity
TSP	MST weight M (OPT > M)	<= 2.0	O(n^2) or O(m log m)
Vertex Cover	Matching size |M| (OPT >= |M|)	<= 2.0	O(nm)
Bin Packing (FF/BF/WF)	sum(s_i)/C (OPT >= sum/C)	<= 2.0	O(n^2)
Bin Packing (NF)	sum(s_i)/C (OPT >= sum/C)	<= 2.0	O(n)
Job Scheduling	sum(t_i)/m and max(t_i)	<= 2.0	O(nm)
Clustering	Virtual (k+1)-th center dist d	<= 2.0	O(k^2 n)

All five problems have approximation ratio **2.0** – giving solutions at most twice the optimal.

Summary

Concept	Definition
P	Problems solvable in polynomial time
NP	Problems whose "Yes" answer is *verifiable* in polynomial time
NP-Hard	At least as hard as every problem in NP
NP-Complete	In NP AND NP-Hard
Reduction $A \leq_P B$	Transform A to B in poly-time, preserving answers
Approximation Ratio	APX / OPT, measured via indirect optimal bound

Key takeaways:

Q & A

39780: Algorithms

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